

ZEYU CHEN

zeyuchen.com
github.com/zecookiez
linkedin.com/in/zeyuchen-dev
zycm03@gmail.com

SKILLS

Languages: Python, C++, C, Java, SQL, JavaScript, Scala, Golang, Ruby, HTML, CSS, Bash

Frameworks & Libraries: OpenCV, NumPy, SciPy, Pandas, Tensorflow / Keras, FoundationDB, PIL, Django, Jinja

Tools: Git, Google Cloud Platform, Google Colab, Datadog, Jupyter Notebook, SageMath

EXPERIENCE

Incoming Software Engineer Intern @ **Snowflake** (September 2023)

Snowflake - Software Engineer Intern

January 2023 - April 2023

Metadata Team

Bellevue, WA

- Architected data allocation monitoring tools using **Java** and **FoundationDB** in a cross-team initiative to prevent corruption on **500 million units** of data tokens generated daily, eliminating over **40 hours** of system downtime
- Achieved over **80%** detection accuracy in abnormal resource usage with an **SQL** alert system using statistical methods
- Scaled account read throughput over **70 times** by overhauling a Java API to introduce a batched query system

Coinbase - Software Engineer Intern

May 2022 - September 2022

Security and Decentralized Identity Team

Remote

- Spearheaded the design of tools to identify security procedure deficiencies by conducting control gap analysis on **100+** security and data governance policy documents with the NIST framework, automating **20%** of the analysis workflow
- Enabled NFT avatar lookup by integrating Coinbase Wallet to the Ethereum Name Service with **Golang**

DMOJ Modern Online Judge - Contest Organizer and Moderator

July 2020 - Present

- Led and co-authored **7+** programming contests averaging **200+** participants each on Canada's largest platform
- Devised graph theory, data structures and dynamic programming problems with published solution instructions

PROJECTS

3D Chess

May 2021 - June 2021

- Designed an autonomous player using **Minimax** search with alpha-beta pruning, **JavaScript**, **WebAssembly** and **C++**
- Reduced the search runtime by over **70 seconds** per turn using pruning methods, heuristics and optimizations

Colonel By Online Judge Programming Platform

July 2020 - Present

- Extended and open-sourced a platform hosted with **Google Cloud Platform** for the school's computer science assessments using **Python**, **Django**, **HTML** and **CSS** — reaching **100+ daily users** and **270+ days** of continuous uptime
- Revamped additional privacy levels with custom point-tallying using **Python** and **MySQL** for student assessments

Mole-croscope (CuHacking 2nd Place, Innovapost Challenge Winner)

Feb 2019

- Created mole anomaly-tracking through object translation invariance with **Python** and **OpenCV** with **84%** accuracy
- Implemented **Python** preprocessing scripts to auto-correct skew and translation differences between user inputs

AWARDS

- Canadian Computing Olympiad (CCO 2021) :: **Silver medalist**
- Canadian Computing Competition (CCC 2021) :: **Perfect score, 1st place, top 1.1%** (~3 000 participants)
- Facebook / Meta Hacker Cup :: **Top 1.2%** in 2020, **top 1.6%** in 2021, **top 1.3%** in 2022 (~30 000 participants)
- Google Kickstart Competition 2021 :: Round C **top 0.4%**, 48th worldwide (~12 200 participants)
- Advent of Code :: **Top 0.03%** in 2020, **top 0.017%** in 2021, **top 0.02%** in 2022 (~260 000 participants)

EDUCATION

University of Waterloo - Bachelor of Software Engineering

Sept 2021 - Present

- 90% CAV and 3.9 / 4.0 GPA — Fall 2021, Winter 2022, Fall 2022 Term Distinction